

Natural Computing

LMU Munich
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Algorithm 6 (basic evolutionary algorithm). Let $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_u \rangle_{0 \leq u \leq t})$ be a population-based optimization process. The process $\mathcal{E}$ continues via an evolutionary algorithm if E has the form

$$E(\langle X_u \rangle_{0 \leq u \leq t}, \tau) = X_{t+1} = \text{selection}(X_t \uplus \text{variation}(X_t))$$

where *selection* and *variation* are possibly randomized or non-deterministic functions so that for any $X \in \wp^*(\mathcal{X})$ it holds that $|\text{selection}(X)| \leq |X|$ and $|\text{selection}(X \uplus \text{variation}(X))| = |X|$.

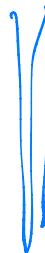
Definition 11 (population-based optimization). Let $\mathcal{X}$ be a state space. Let $\mathcal{T}$ be a target space with total order $\leq$. Let $\tau : \mathcal{X} \rightarrow \mathcal{T}$ be a target function. A tuple $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_t \rangle_{0 \leq t \leq g})$ is a population-based optimization process iff $X_t \in \wp^*(\mathcal{X})$ for all t and $E : \langle \wp^*(\mathcal{X}) \rangle \times (\mathcal{X} \rightarrow \mathcal{T}) \rightarrow \wp^*(\mathcal{X})$ is a possibly randomized, non-deterministic, or further parametrized function so that the population-based optimization run is produced by calling E repeatedly, i.e., $X_{t+1} = E(\langle X_u \rangle_{0 \leq u \leq t}, \tau)$ where X_0 is given externally or chosen randomly.

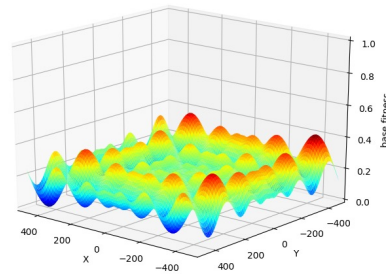
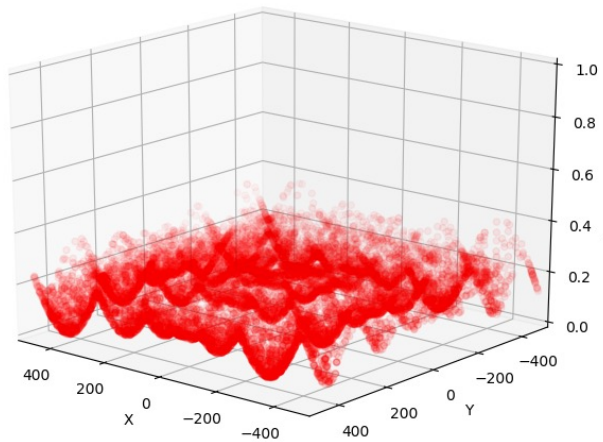
$$x_0 \sim \mathcal{X}$$

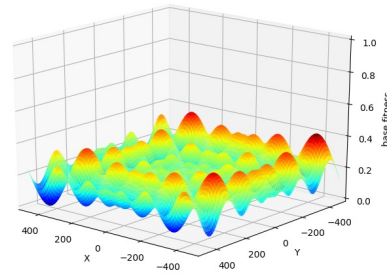
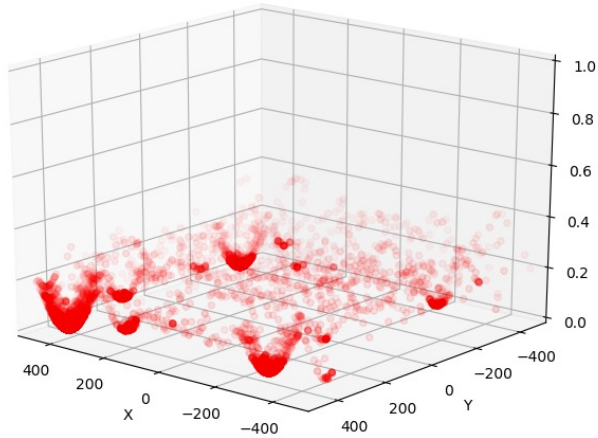
$$X_0 \sim \wp^*(\mathcal{X})$$

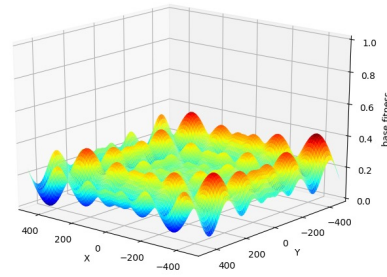
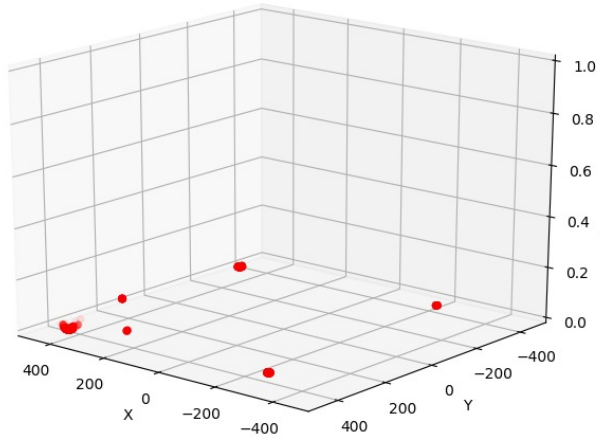
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Definition 12 (population-based optimization (alternate)). An optimization process $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_t \rangle_{0 \leq t \leq g})$ is called population-based iff $\mathcal{X}$ has the form $\mathcal{X} = \wp^*(\mathcal{Y})$ for some other state space $\mathcal{Y}$.









Variation



very small
adaptations



larger partial
"good" solutions
combined

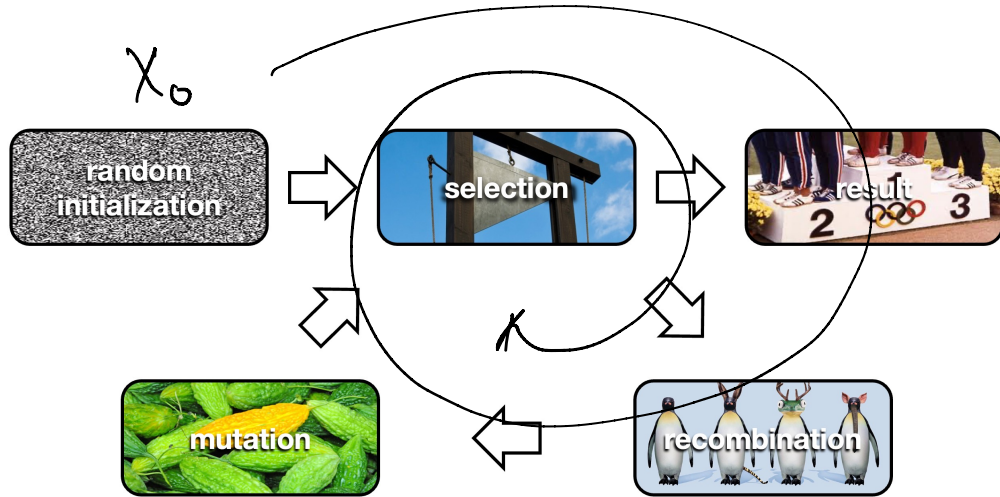


image sources:

www.bostonmagazine.com/news/2015/07/30/boston-2024-winners-losers

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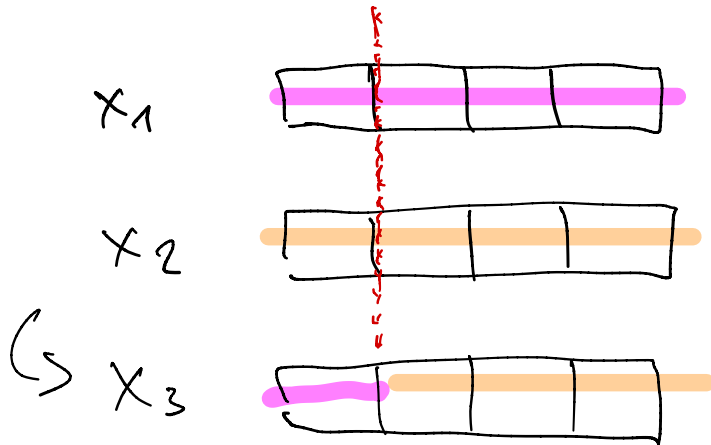
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<https://phys.org/news/2019-10-guillotine-cruel-poisoning.html>

Recombination

1 point $\sim$ crossover $\sim$

$$\mathcal{X} * \mathcal{X} \rightarrow \mathcal{X}$$



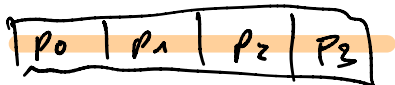
$$\mathcal{X} \in \mathbb{R}^4$$

$$\begin{aligned} & [: \text{cop.}]^\# \\ & \approx [\text{cop} :] \end{aligned}$$

Mutation

$$\mathcal{X} \rightarrow \mathcal{X}$$

x_i



$$x \in \mathbb{R}^h$$

x_i'



$$r_i \sim [0 \dots 4]$$

$$= 1$$

$$r_j \sim [0; 1]$$

Migration/Hypermutation

$$\emptyset \rightarrow \mathcal{X}$$

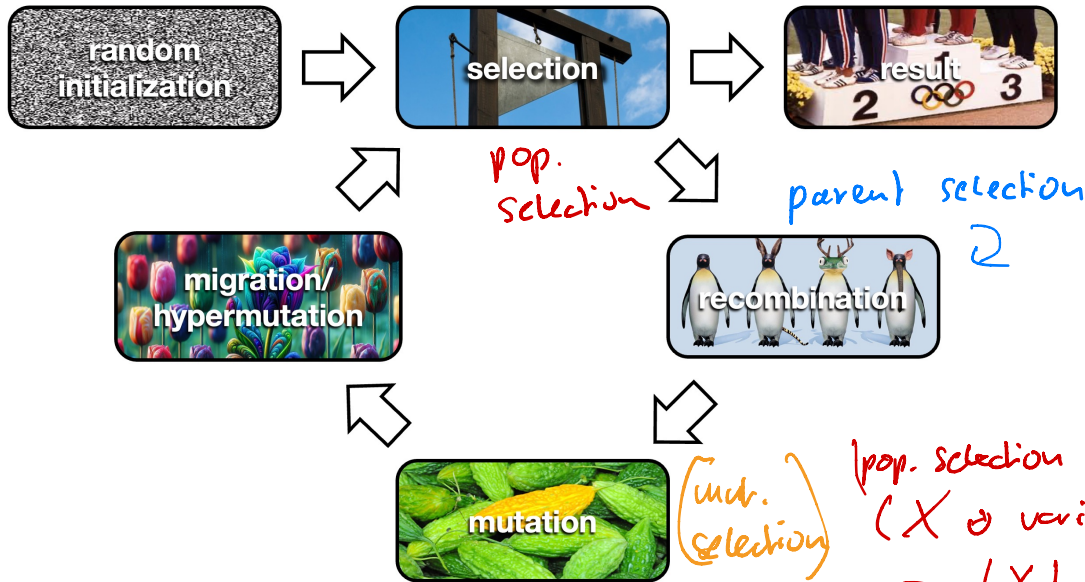


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Natural vs. Artificial Selection

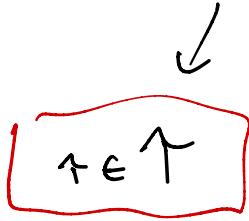


too complex,
many correlating factors,
stochastic

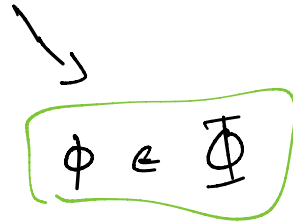


very abstract,
but useful
for optimization

Target Function vs. Fitness Function


$$\tau \in \mathbb{T}$$

the optimization
target
(numerical)


$$\phi \in \Phi$$

$$\phi \preceq \tau \quad \textcircled{\checkmark}$$

$$\phi \preceq \tau + \text{_____}$$

$$\tau = \begin{cases} \tau, & \phi \text{ "good"} \\ \text{_____} & \text{otherwise} \end{cases}$$

Points of Selection

pop. selection

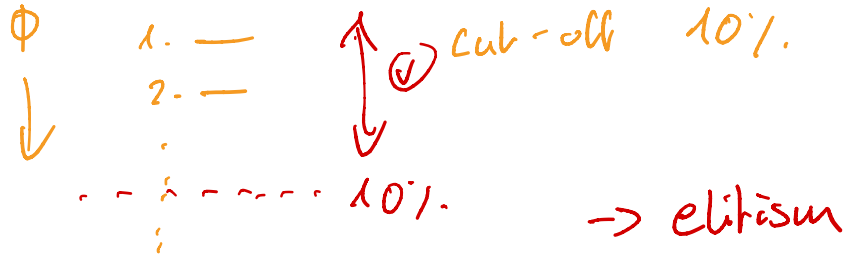
parent selection

(mutation selection)

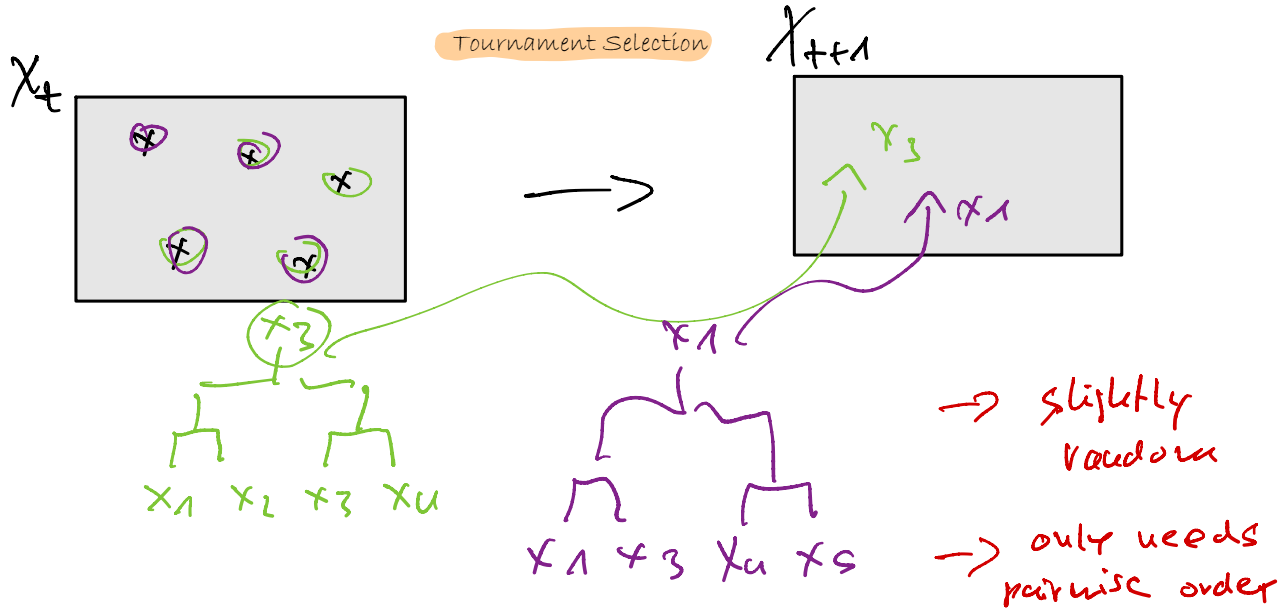
Types of Selection

Cut-Off Selection

↳ "take the first n good samples"

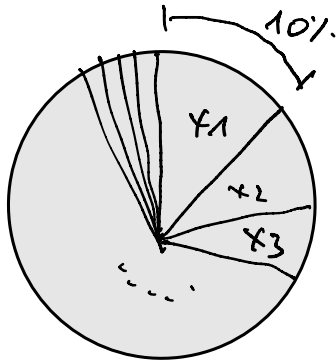


Types of Selection



Types of Selection

Roulette Wheel Selection



$$\sum_{i=0}^{|X|-1} \phi_i$$

$$\phi(x) = 10\%$$

proportionally
→ stochastic

↑ $r \sim [0; 1]$

Algorithm 7 (typical evolutionary algorithm). Let $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \phi, E, \langle X_u \rangle_{0 \leq u \leq t})$ be a population-based optimization process.

- Let $\sigma_N^{\text{survivors}}, \sigma_N^{\text{parents}} : \wp^*(\mathcal{X}) \rightarrow \wp^*(\mathcal{X})$ be (possibly randomized or non-deterministic) selection functions that return $N \in \mathbb{N}$ individuals, i.e., $|\sigma_N(X)| = N$ and $\sigma_N(X) \subseteq X$ for all X . They may possibly depend on $\mathcal{E}$ (most commonly ϕ). Let $\varrho_N^{\text{mutants}} : \wp^*(\mathcal{X}) \rightarrow \wp^*(\mathcal{X})$ be a special selection function that, given a population X , returns a random subset $X' \subseteq X$ so that $|X'| = N$. Note that $\varrho_{|X|}(X) = X$ for all X .
- Let $\text{mutate} : \mathcal{X} \rightarrow \mathcal{X}$ be a (possibly randomized or non-deterministic) single-individual mutation function. We write $\text{mutate}[\cdot] : \wp^*(\mathcal{X}) \rightarrow \wp^*(\mathcal{X})$ for the per-individual application of mutate , i.e., $\text{mutate}[X] = \{\text{mutate}(x) : x \in X\}$.
- Let $\text{recombine} : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{X}$ be a (possibly randomized or non-deterministic) two-parent recombination function. We write $\text{recombine}[\cdot] : \wp^*(\mathcal{X}) \rightarrow \wp^*(\mathcal{X})$ for the random-pairing application of recombine , i.e., $\text{recombine}[X] = \{\text{recombine}(x_1, x_2)\} \uplus \text{recombine}[X \setminus \{x_1, x_2\}]$ with $x_1, x_2 \sim X$ and $\text{recombine}[\{x\}] = \emptyset$ for all x as well as $\text{recombine}[\emptyset] = \emptyset$.
- Let $\text{migrate} : \emptyset \rightarrow \mathcal{X}$ be a (possibly randomized or non-deterministic) individual creation function. We write $\text{migrate}_N : \emptyset \rightarrow \wp^*(\mathcal{X})$ with $N \in \mathbb{N}$ for the iterated application of migrate , i.e., $\text{migrate}_N[\cdot] = \{\text{migrate}()\} \uplus \text{migrate}_{N-1}[\cdot]$ with $\text{migrate}_0[\cdot] = \emptyset$.

The process $\mathcal{E}$ continues via a typical evolutionary algorithm if E has the form

$$\begin{aligned} E(\langle X_u \rangle_{0 \leq u \leq t}, \phi) = X_{t+1} = & \sigma_{|X_t|}^{\text{survivors}}(X_t \\ & \uplus \text{mutate}[\varrho_M^{\text{mutants}}(X_t)] \\ & \uplus \text{recombine}[\sigma_{2C}^{\text{parents}}(X_t)] \\ & \uplus \text{migrate}_H[\cdot]) \end{aligned}$$

where $M \in \mathbb{N}$ is the number of mutants produced per generation, $C \in \mathbb{N}$ is the number of children produced per generation, and $H \in \mathbb{N}$ is the number of migrants (sometimes called hypermutants) generated per generation. Especially M is often expressed as a rate for random choice within the population, i.e., $M = \lceil m \cdot |X_t| \rceil$ where $m \in \mathbb{R}$ is called mutation rate.

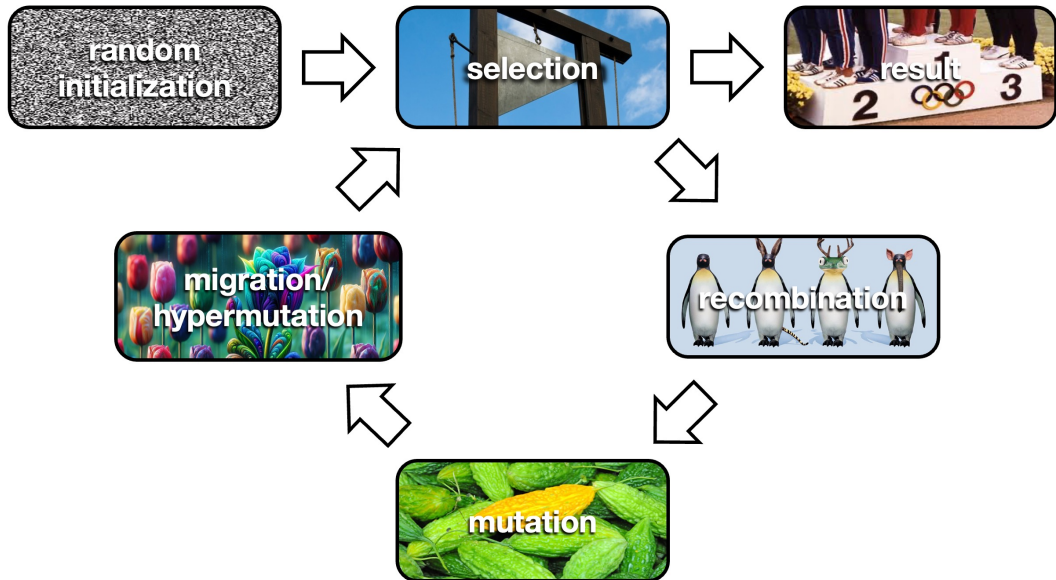


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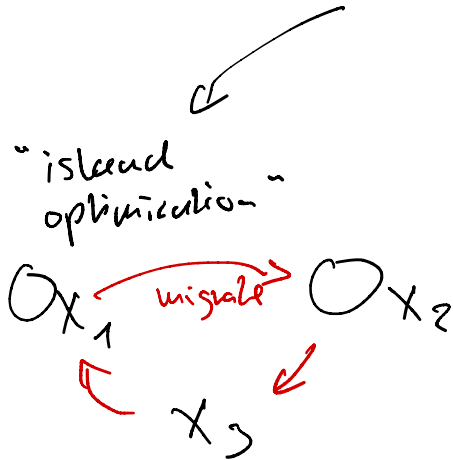
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Exploration/Exploitation for EAs

→ Lots of options



↓
all the
hyperparameters

→ ...
see
liberbur

Diversity

$$\phi(x) = f(x) + \frac{\quad}{\quad}$$

↑
insert
diversity term
here

→ to avoid premature
convergence in
populations

- 'distance'
of samples
to each
other

→ penalize "similarity"

- "novelty"

→ "different" solutions are desirable

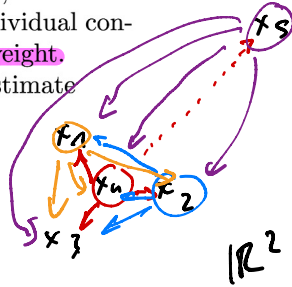
Algorithm 8 (diversity-aware evolutionary algorithm). Let $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \phi, E, \langle X_u \rangle_{0 \leq u \leq t})$ be the process of an evolutionary algorithm (cf. Alg. 7). $\mathcal{E}$ is called a diversity-aware evolutionary algorithm if the used fitness function $\phi : \mathcal{X} \rightarrow \mathcal{T}$ has the form

$$\phi(x) = (1 - \zeta) \cdot \tau(x) + \zeta \cdot \delta(x)$$

where $\tau : \mathcal{X} \rightarrow \mathcal{T}$ is the underlying target function of the given optimization problem, $\delta : \mathcal{X} \rightarrow \mathcal{T}$ is a (possibly randomized, non-deterministic, or further parametrized) function that estimates the diversity that a given individual contributes to the population, and $\zeta \in [0; 1] \subset \mathbb{R}$ is called a **diversity weight**. Note that according to Thm. 4 and Gabor et al. [2] the diversity estimate

$$\delta(x) = \frac{1}{|X_t|} \cdot \sum_{x' \in X_t} |x' - x|$$

is generally suitable for $\mathcal{X} = \mathbb{R}^n$ or $\mathcal{X} = \mathbb{Z}^n$ for some n .



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(2003)

Theorem 4 (Wineberg-Oppacher diversity [4]). All measures of the diversity of an individual x in a population X are related to the pairwise distance between individuals in X as measured by a suitable norm (Euclidean for real-valued representations, Hamming for symbolic representations, e.g.).

The Ideal Fitness Function?

↳ doesn't exist (yet)